unit-1

Definition of operation research.

The term operation research as application of scientific method to problem arising from operation involving men, machines and material it normally utilizes the knowledge and skill of an inter disciplinary research team to provide the managers of search system with optimum operating solution

characteristic of OR

The essential characteristic of OR are:

1. inter disciplinary team approach: the highly optimized solution is find by a group of scientist who are selected from different field.

2. To Maximize the total output: It attempts to maximize the total output by maximizing the profit and minimizing the loss.

3. use of scientic research: It makes use of scientific research to arrive at optimum solution

4 Wholastic approach to the system: operation reseach takes into account all significant related factor and finds the best optimum solution for the total organisation

need of OR

The purpose of operation research is to use strategies and war like methods to kill the enemy and rise victories in the battle of business.

The need for operation research is felt on the account of following reason

1. complex business condition such as customer demand, availability raw material, equipment, capacity etc
2. scattered responsibility and authority
3. uncertain business environment
4. knowledge explosion

scope of OR / application

operation research as wide scope in almost in all the areas where there is a need for optimality also for optimum distribution of available resources so as to ensure maximum profit. operation research applicability of the following areas.

1. industrial management
2. Defence operation
3. Hospital management
4. Transport management
5. B...

b. Economic planning

Work of OR.

The working of operation research involves the following step

1. Formulation of the problem, identifying the objectives and decision variable and also the constraints that arise involving the decision variable

2. Construction of mathematical model expressing the measure of effectiveness which may be total profit, total cost, utility etc to be obtimized by a mathematical function called objective function representing the constraints like budget constraints raw material constraints resource constraints, quality constraint etc.

3. Solving the model constructor finding the solution by analytic or interactive method depending upon the structure of the mathematical model

4. Establishing control over the solution for the purpose, adequate control must be over the variables in order to retain their continuous and non-changing character.

5. Testing the model and its solution checking as far as possible either from the past available data or by experience

wheatne the model gives solution which can be used in practise

6. implementation : implement using the solution to achive the desire goal

operation research techniques

Following are the varies techniques of operation research

1. Linear programming : Linear programming is a mathematical techniques of assigning such a fixed amount of resources to satisfy a number of demands in way some objective is optimize and other define condition also satisfy

2. Transportation problem : The transportation problem is a special type of LPP where the objective is to minimize the past of distribution from a number of sources to number of destination

3. Assignment problem : When the problem involves the allocation of different facilities to different task, it is termed as an assignment problem.

4. Queuing theory : The queuing problem is identify the presence of a group of customer to arrive randomly receive some service

this theory helps in calculating the expected number of people in the queue, expected idle item of the server etc.

5. Game theory: It is used for decision making under confidencity situation when one or more components in game theory. The game theory provides solution to the game assuming that each of that search game players wants to maximize this profit and minimize this loss.

6. inventory control model: It is concern with the storage, handeling of goods so as to ensure the availability of stock whenever needed and minimize the wastage and losses

7. simulation: It is a technique that involves setting up a model of a real situation and then performing experiment

8. Decision theory: This help better decision making under the condition of uncertainity

9. Margo analysis: this make analysis of current behaviour of some variable to predict the future behaviour of variable

Limitations

operation research as an emerging field of managerical sciente. inspite of thi it suffers from the following drawbacks

1. problem in formulation : operation research is built on the basis of problem formulation . formulating problem to find optimal solution posses much difficulties

2. Difficulties in Data Collection : operation research involves around the use of information for decision making in actual practice , collection of quality data at the right time is often difficult .

3 Difficulties of findind solution on time : Due the complex nature of problem solving excersise , most often it is difficult to find optimum solution on time

4. Resistance from employees : operation research helps to find solution for many of ~~tuog~~ tough problems faced by the management there may be resistance in this approach

5. Uses of computers : Many of the problem involving operation research are so complex that they can be solved only with the help of computer. Even after finding optimal solution it is rather difficult for the managa to find the outcome and the result of the optimal solution . This is an important reason why many times the use of operation

research technique fails to get support from the administration

Application of OR

operation research is applicable in the following area of

1. Accounting & Finance : In the field of accounting operation research is helpful in cashflow planning and credit policy analysis. Further operation research is useful in planning strategy for accounting operation research is helpful in capital budgeting decision, portfolio decision, dividend decision etc.

2. Marketing : operation research is useful in selection of product mix and advertising and media planning, travelling salesman decision etc.

3. Human resources : operation research plays an important role on various aspect of requirement and selection of personal. it is used in designing of the mix of age and skills, man power planning, salary administration, assignment of job etc.

4. Manufacturing : operation research is useful in various area of manufacturing including production planning, scheduling,

inventory control etc

5. procurement : operation research is useful in various areas of purchasing on the procurement such as vendor analysis, replacement decision

6. project : operation research helps in successfully project management in various area of project scheduling, maintaining schedule ; allocation of resource etc

7. logistic : operation research is useful in various area of ~~manufactu~~ logistic and supply chain management this includes decision involving location and size of warehouse, transportation etc.

Linear programming

Definition

Linear programming is the analysis of problem in which a linear function of a number of variable is to be optimize when those variables are subject to a number of constraints in mathematical linear inequalities.

General form

The linear programming involved in more than two variable may be expressed as follows

maximize (or) minimize $Z = c_1 x_1 + c_2 x_2 + c_n x_n$

subject to the constraint

$a_{11} x_1 + a_{12} x_2 + \cdots + a_{1n} x_n \leq \text{or} = \text{or} \geq b_1$

$a_{21} x_1 + a_{22} x_2 + \cdots + a_{2n} x_n \leq \text{or} = \text{or} \geq b_2$

$\cdots \cdots \cdots \cdots$

$a_{m1} x_1 + a_{m2} x_2 + \cdots + a_{mn} x_n \leq \text{or} = \text{or} \geq b_m$

and the non-negative restriction

$x_1, x_2 \cdots x_n \geq 0$

Objective function

The linear function $Z = c_1 x_1 + c_2 x_2 + \cdots + c_n x_n$ which is to be minimize or maximize is called the objective function

constraints

The are always certain limitation on the use of limited resource this limitation are called as constraints

Decision variable

The variable $x_1, x_2 \cdots x_n$ involved in the objective function are called decision variable.

linear programming composation

procedure for forming the LPP model

step 1 - identify the unknown decision variable to be calculation and assign symbol to them

step 2 - identify on the restriction or constraint in the problem and express them as linear equation or inequalities of decision variable

step 3 - identify the aim or objective and represent it also as the linear function of decision variable.

step 4 - Express the complete formulation of LPP as a general mathematical model.

Assignment problem

Definition

Assignment perobem is a special case of transportation problem in which the objective is to assign a number of resources to an equal number of activities so as to minimize the total cost or maximize the total profit of the allocation

ex: a number of men assigned with equal number of jobs

The assignment problem can be stated in form of $n \times n$ matrix (c_{ij}) is called a cost matrix or effectiveness matrix where c_{ij} is the cost of assigning ith machine to the jth job.

Unbalanced assignment problem

If the number of rows is not equal to the number of columns in the cost matrix then given assignment problem is said to be unbalance

Mathematical formulation of an assignment problem

20/7/22

Linear programing problem => LPP

Linear - Degree power on

$\Downarrow$

use of resource.

less than $\le$ Maximize
greater than $\ge$ Minimize } dual

1. kavin Ltd produces and sells two different product under the brand name black and white respectively. The profit per unit on these product is ₹ 50 and ₹ 40 respectively both the product employeed the same manufacturing process, which has a 50000 man hours; as per the estimate the market demand for maximum 8000 units of black and 10000 units of white. Subject to the overall demand the product can be sold in any possible constraints if it takes 3 hours to produce 1 unit of black and 2 hours to produck 1 unit of white. Formulate the model of linear programing problem

sol x_1 - be the no. of units produced in black

x_2 - be the no. of units produced in white.

maximize $Z = 50x_1 + 40x_2$

St => subject to the constraint

$$3x_1 + 2x_2 \le 50000$$
$$x_1 \le 8000$$
$$x_2 \le 10000$$
$$x_1, x_2 \ge 0$$

2 A firm produces 3 product are process on 3 different machines, time require to manufacture 1 unit of each three product and daily capacity of three machines are given in the table.

Time / unit				M/c capacity maximum / day
M/c	P_1	P_2	P_3	
M_1	2	3	2	440
M_2	4	-	3	470
M_3	3	5	-	430

the profit per unit for product 1 2 3 or 4 3 6 respectively. Formulate the LPP

sol x_1 - be the no. of units produced in P_1

x_2 - be the no of units produced in P_2

x_3 - be the no. of units produced in P_3

$$\text{MAX } Z = 4x_1 + 3x_2 + 6x_3$$

ST

$$2x_1 + 3x_2 + 2x_3 \le 440$$

$$4x_1 + 0x_2 + 3x_3 \le 470$$

$$3x_1 + 5x_2 + 0x_2 \le 430$$

$$x_1, x_2, x_3 \ge 0$$

3 A firm manufacture 2 types of product

A and B sells them at profit ₹ 2 and 3 each product process on 2 machine type A require 1 minute of processing time machine 1 and 2 minute of processing time machine 2, type B requires 1 minute of machine 1 and 1 min of machine 2 machine 1 available not more than 6 hours and 40 min. Machine 2 available for 10 hours formulate LPP

sol x_1 - be the no. of units produced in type A

x_2 - be the no. of units produced in type B

$$Max\ z = 2x_1 + 3x_2$$

	M_1	M_2
A	1 min	2 min
B	1 min	1 min
	6 hrs 40 min	10 hrs
	360 + 40	10 × 60
	400 min	600 min

ST

$$x_1 + x_2 \leq 400 min$$

$$2x_1 + x_2 \leq 600 min$$

$$x_1, x_2 \geq 0$$

---

4 A Dietition wishes to mixes two types of fruit in such a way that the vitamin constrain of the

mixture contain atleast 8 unit of vitamin A and 10 units of vitamin B. Food 1 contain 2 unit per kg of vitamin A and 1 unit per kg of vitamin B While Fruit 2 contain 1 unit per kg vitamin A and 2 unit per kg of vitamin B. If cost Rs. 5 per kg to purchase fruit 1 and Rs. 8 per kg to purchase fruit 2. Formulate the LPP

sol x_1 - be the no. of kg produced in Food Fruit 1

x_2 - be the no. of kg produced in Food Fruit 2

Minimize $z = 5x_1 + 8x_2$

ST

vit A	vit B
2	1
1	2
8	10

ST

$2x_1 + x_2 \geq 8$

$x_1 + 2x_2 \geq 10$

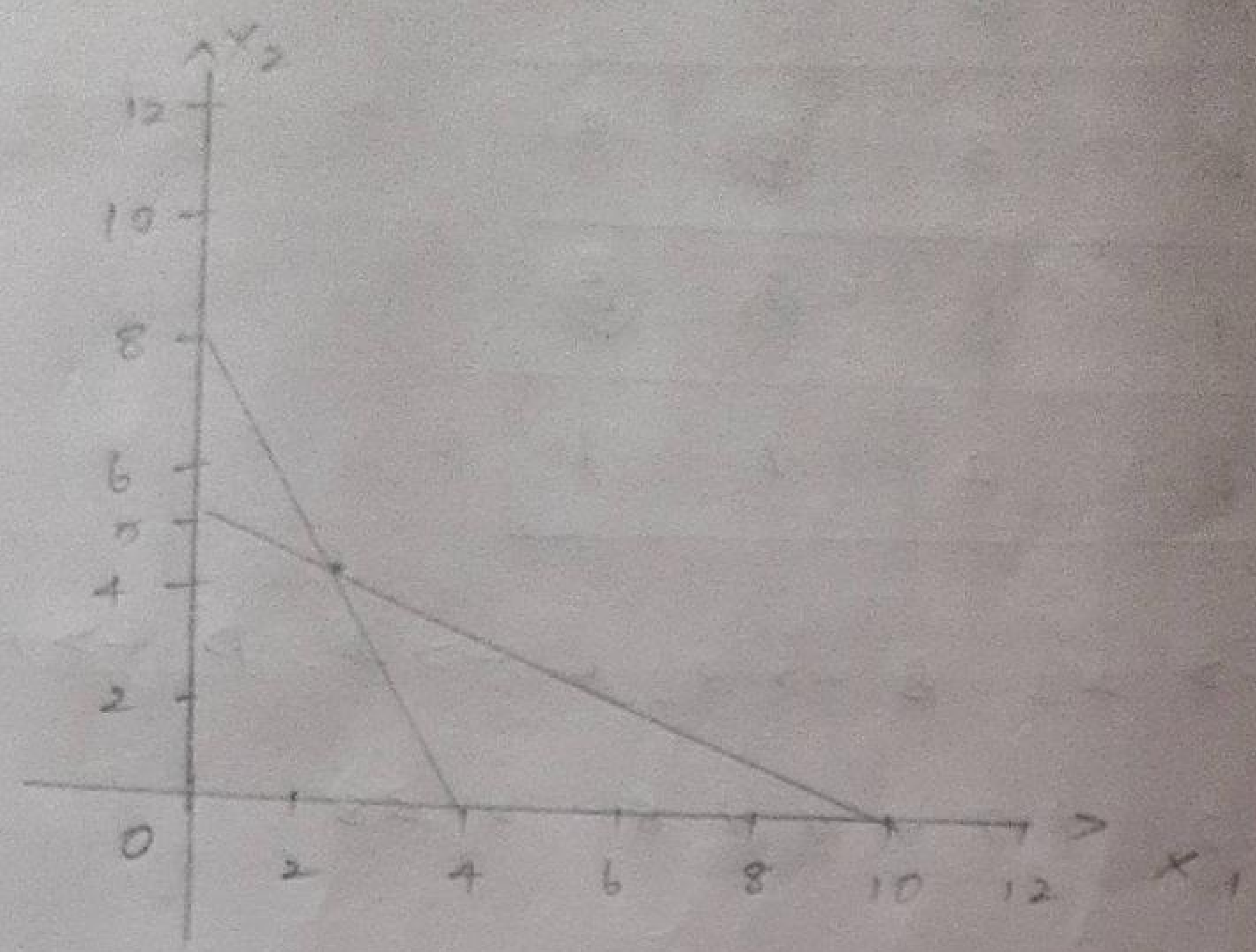
$x_1, x_2 \geq 0$

Graphical method

$2x_1 + x_2 = 8$ $x_1 + 2x_2 = 10$

Scanned by TapScanner

x_1	0	8		x_1	0	10
x_2	4	0		x_2	5	0



$x_1 = 2$, $x_2 = 4$

25/7/22

Assignment problem => resources allocat to the job

1.

	1	2	3	4
A	10	12	19	11
B	5	10	7	8
C	12	14	13	11
D	8	12	11	19

-> Jobs

Machine

cost

Row reduction

0	2	9	1
0	5	2	3
1	3	2	0
0	4	3	1

column reduction

0	0	7	1
0	3	0	3
1	1	0	0
0	2	1	1

Allocation

~~0~~	0	7	1
~~0~~	3	0	3
1	1	~~0~~	0
0	2	1	1

A -> 2, B -> 3, C -> 4, D -> 1

12 + 7 + 11 + 8 = 38

Row red

Step 1 - First we have check no. of row equal to no. of columns.

Step 2 - Row reduction - Find the least cost in every row and deduct the row elements.

Step 3 - column reduction - Find the least cost in every column and deduct the corresponding column.

Step 4 - Find the unique zero in the matrix and encircle the zero, is there any other zero's in corresponding row and column eleminate those zero.

Step 5 - Repeat the step 4 untill the all row got allocation step 6 - No. of allocation is equal to No. of row or column, that is we reach the optimum solution for the problem

Minimize z = cost of the allocation.

2. Find the optimum Assignment problem solution

	W	X	Y	Z
A	11	17	8	16
B	9	7	12	6
C	13	16	15	12
D	14	10	12	11

Sol Row reduction

3	9	0	8
3	1	6	0
1	4	3	0
4	0	2	1

column reduction

2	9	0	8
2	1	6	0
0	4	3	0
3	0	2	1

2	9	0	8
2	1	6	0
0	4	3	~~0~~
3	0	2	1

A -> Y, B -> Z, C -> W

D -> X

8 + 6 + 13 + 10

= 37

3

	1	2	3	4	5
A	13	8	16	18	19
B	9	15	24	9	12
C	12	9	4	4	4
D	6	12	10	8	13
E	15	17	18	12	20

Sol Row reduction

5	0	8	10	11
0	6	15	0	3
8	5	0	0	0
0	6	4	2	7
3	5	6	0	8

column reduction

5	0	8	10	11
0	6	15	0	3
8	5	0	0	0
0	6	4	2	7
3	5	6	0	8

	1	2	3	4 ✓	5	
A	5	0	8	10	11	
B	~~0~~	6	15	~~0~~	3	✓
C	8	5	0	~~0~~	~~0~~	
D	0	6	4	2	7	
E	3	5	6	0	8	✓

	1	2	3	4	5
A	5	0	8	10	11
B	0	3	12	0	0
C	8	5	0	0	0
D	0	6	4	2	7
E	0	2	3	0	5

A -> 2, B -> 5, C -> 3, D -> 1, E -> 4

8 + 12 + 4 + 6 + 12 = 42

4 Find the optimum cost for Assignment 4 x 4 matrix

	M_1	M_2	M_3	M_4
A_1	5	7	11	6
A_2	8	5	7	6
A_3	4	7	10	7
A_4	10	4	8	3

sol

Row reduction

	M_1	M_2	M_3	M_4
A_1	0	2	6	1
A_2	3	0	4	1
A_3	0	3	6	3
A_4	7	11	5	0

column reduction

	M_1	M_2	M_3	M_4
A_1	0	2	2	1
A_2	3	0	0	1
A_3	0	3	2	3
A_4	7	1	1	0

Allocation

	M_1	M_2	M_3	M_4
A_1	0	1	1	0
A_2	3	0	0	1
A_3	0	2	1	2
A_4	7	1	1	0

	M_1	M_2	M_3	M_4
A_1	0	0	0	0
A_2	3	0	0	1
A_3	0	2	1	2
A_4	6	0	0	0

$A_1 \to M_3$, $A_2 \to M_2$, $A_3 \to M_1$, $A_4 \to M_4$

$11 + 5 + 4 + 3 = 23$

5. Form the Assignment problem

	1	2	3	4	5
A	3	8	2	10	3
B	8	7	2	9	7
C	6	4	2	7	5
D	8	4	2	3	5
E	9	10	6	9	10

If the number of allocation less than no. of row or column

A: Row reduction

	1	2	3	4	5
A	1	6	0	8	1
B	6	5	0	7	5
C	4	2	0	5	3
D	6	2	0	1	3
E	3	4	0	3	4

column reduction

	1	2	3	4	5
A	0	4	0	7	0
B	5	3	0	6	4
C	3	0	0	4	2
D	5	0	0	0	2
E	2	2	0	2	3

0	2	~~0~~	5	~~0~~
5	3	0	6	4
1	0	~~0~~	2	~~0~~
3	~~0~~	~~0~~	0	~~0~~
2	2	~~0~~	2	3

0	1	~~0~~	4	~~0~~
5	3	0	6	4
~~0~~	0	~~0~~	1	~~0~~
2	~~0~~	~~0~~	0	~~0~~
2	2	~~0~~	2	3

=> Step 1 : Select the unallocated row is there any zero mark the column, is there any allocation in that column mark that row.

step 2 : strike out unmarked rows and marked column [this step is for cover all the zero].

step 3 : Find the minimum cost from uncovered element reduce the cost from the element.

step 4 : Repeat the step 3 . untill reach the optimum allocation.